Applications and extensions - Meta-analytic Structural Equation Modeling (MASEM)

G0B75A Meta-analysis

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Outline

- $\boxed{\text{SEM}}$
    - An example
- MASEM
    - An example

Structural Equation Modeling (SEM)

- Consist of a **structural model** and/or a **measurement model**, that are interconnected and estimated simultaneously
  - The **structural model** describes relationships between multiple variables
  - A **measurement** model is used to link latent (=unobserved variables) to manifest (=observed) variables

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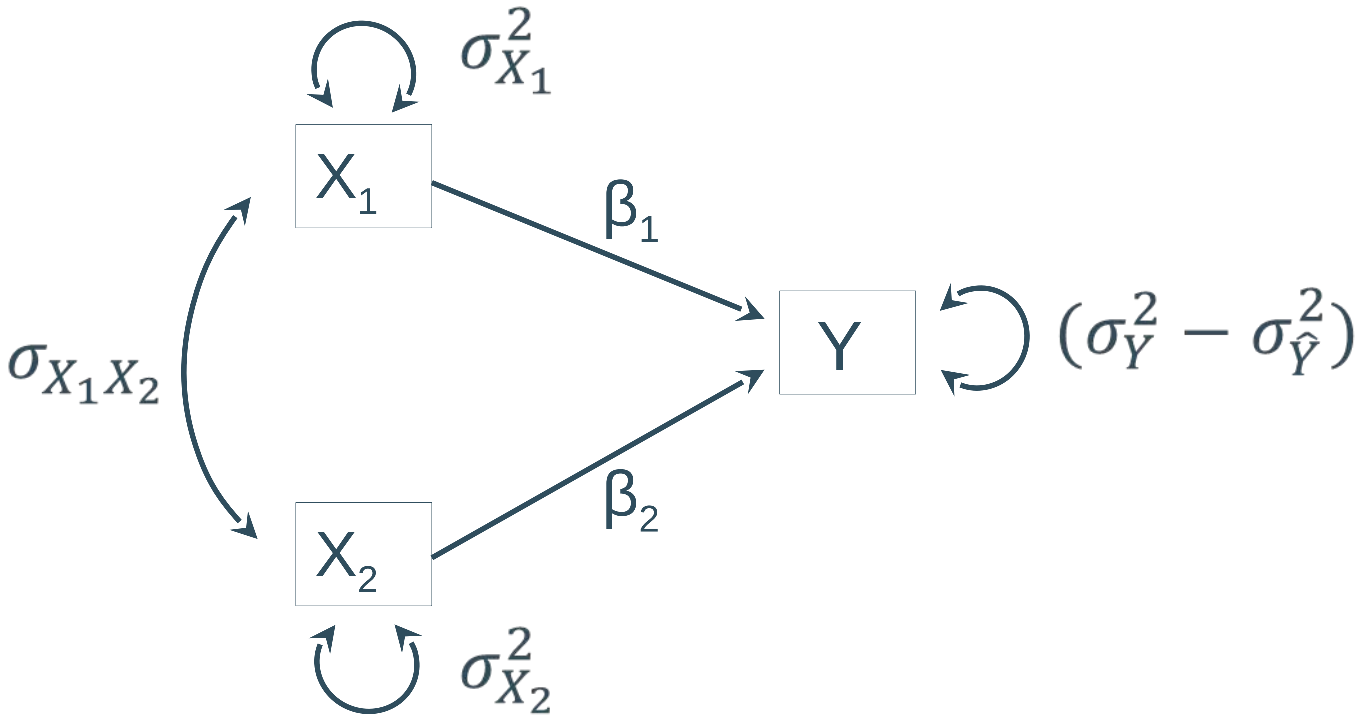
  - The **structural model** describes relationships between multiple variables
  - A **measurement** model is used to link latent (=unobserved variables) to observed variables

Structural Model

Examples of **structural models**:

- multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_1 X_{1i} + e_i$$



Structural Model

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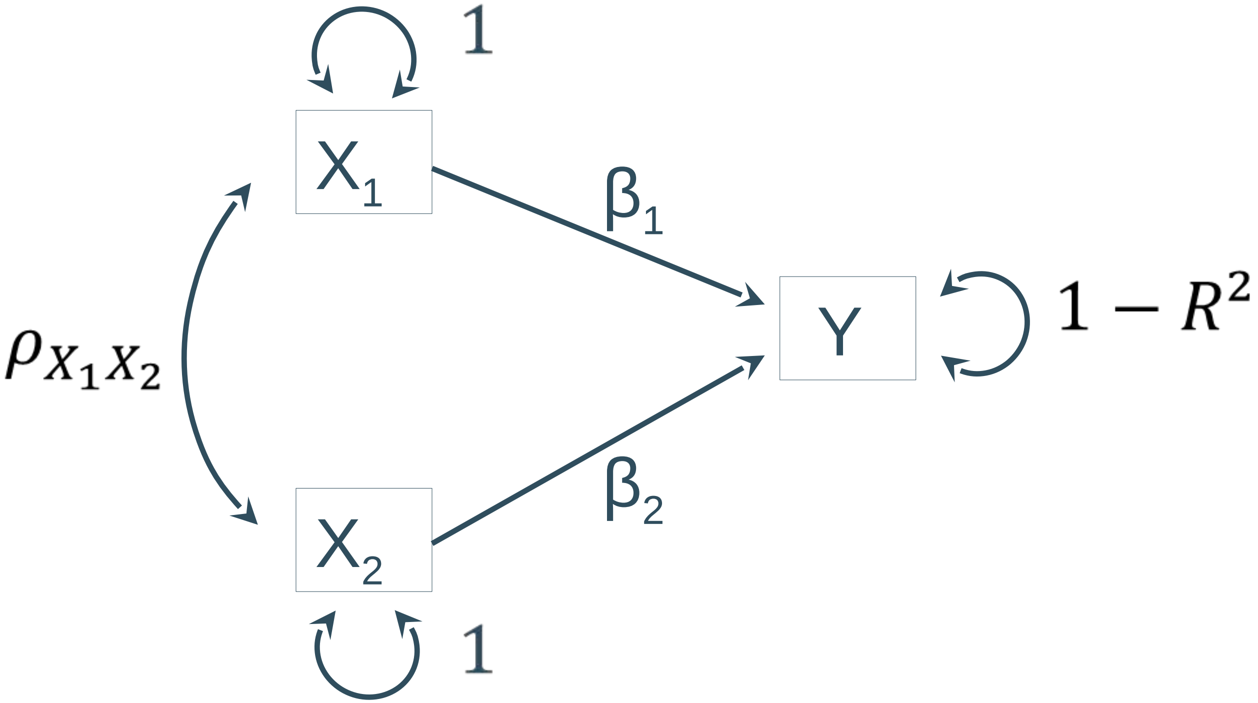
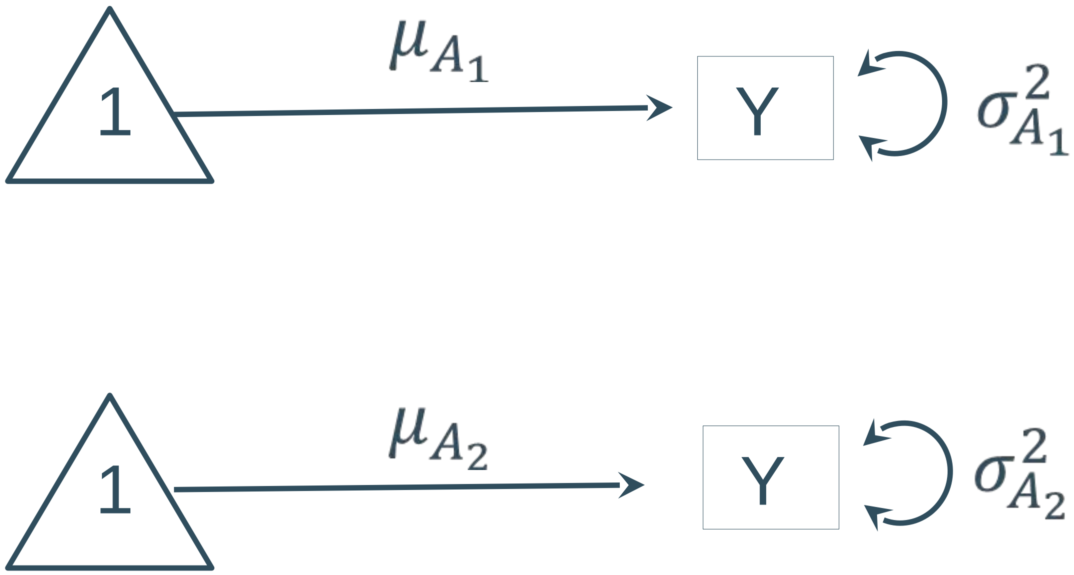


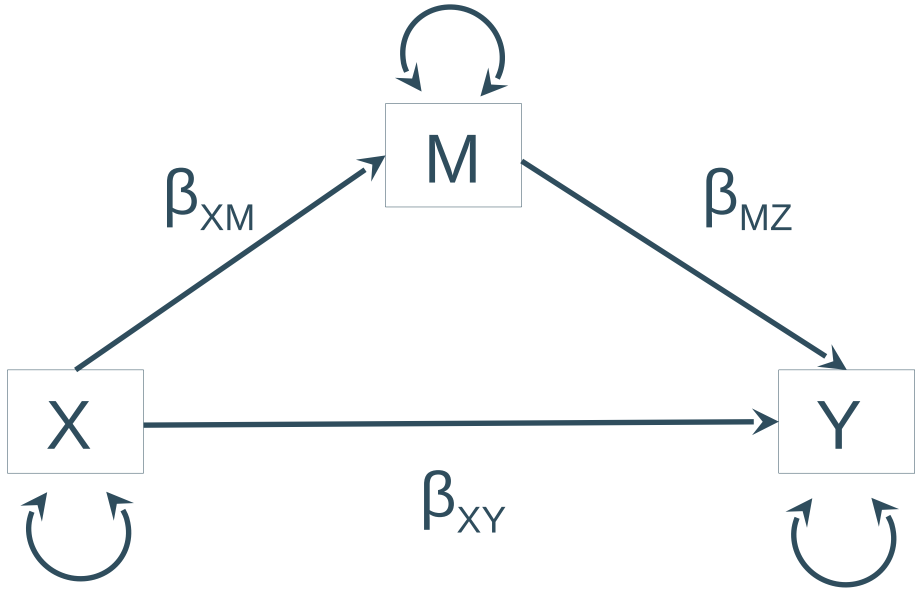
Structural Model

- Examples of **structural models**:
  - ANOVA model



Structural Model

- Examples of **structural models**:
  - mediation model

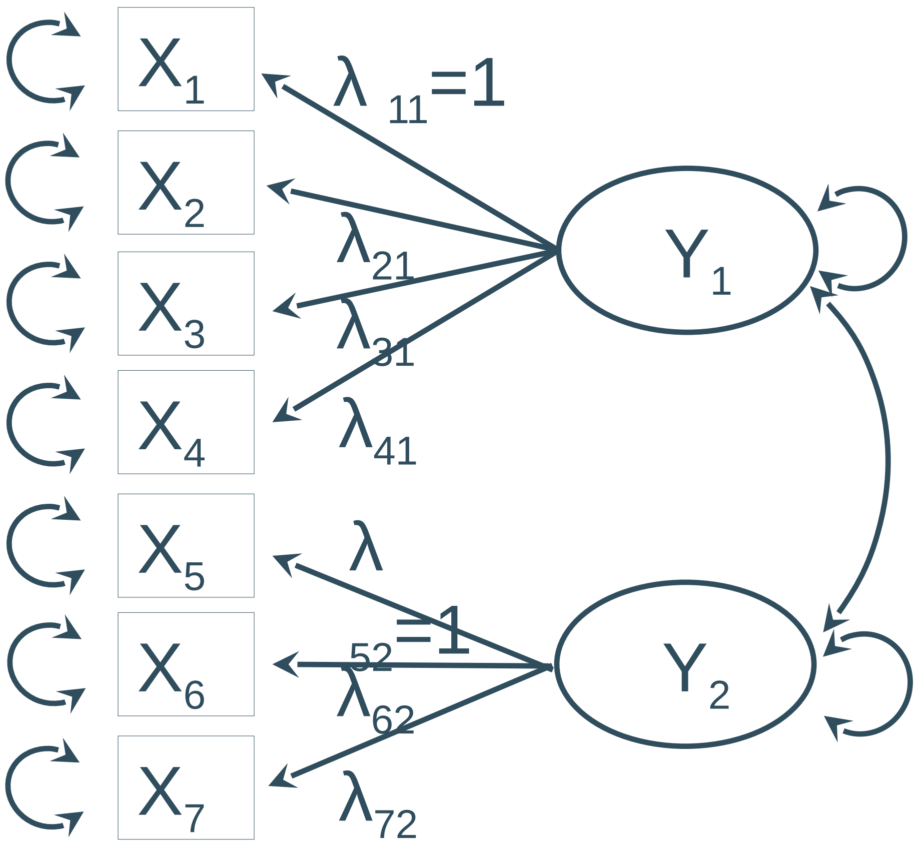


Structural Equation Models

- Consist of a **structural part** and/or a **measurement part**, that are interconnected and estimated simultaneously
  - The **structural model** describes relationships between multiple variables (Cfr. path models)
  - A **measurement** model is used to link latent (=unobserved variables) to observed variables

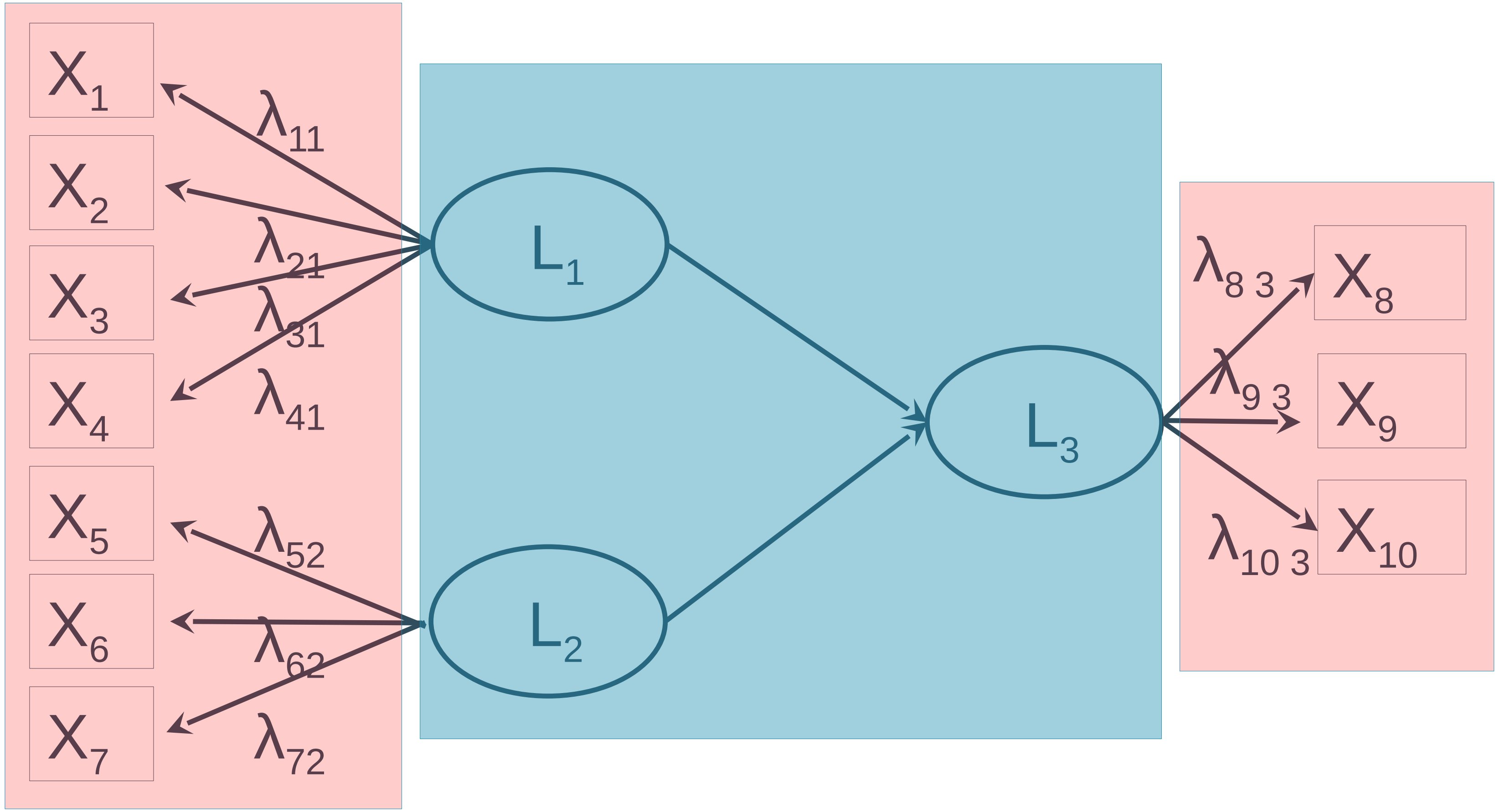
Measurement model

- Examples of **measurement models**:
  - Confirmatory factor analysis model



Structural Equation Models

- **Combining** structural and measurement models (to a latent variable structural model)



Outline

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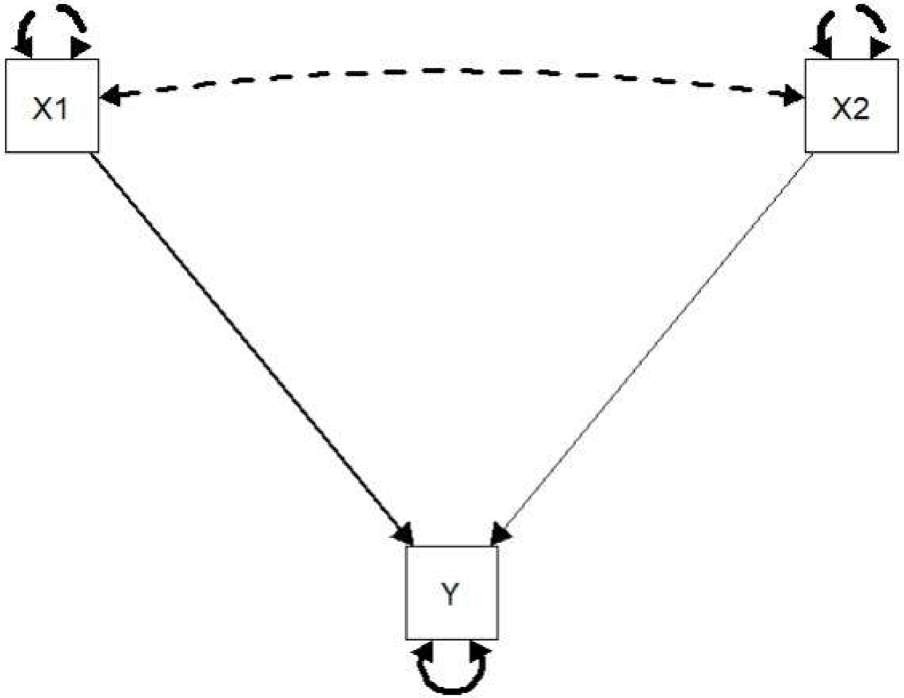
- MASEM
  - An example

SEM: An example

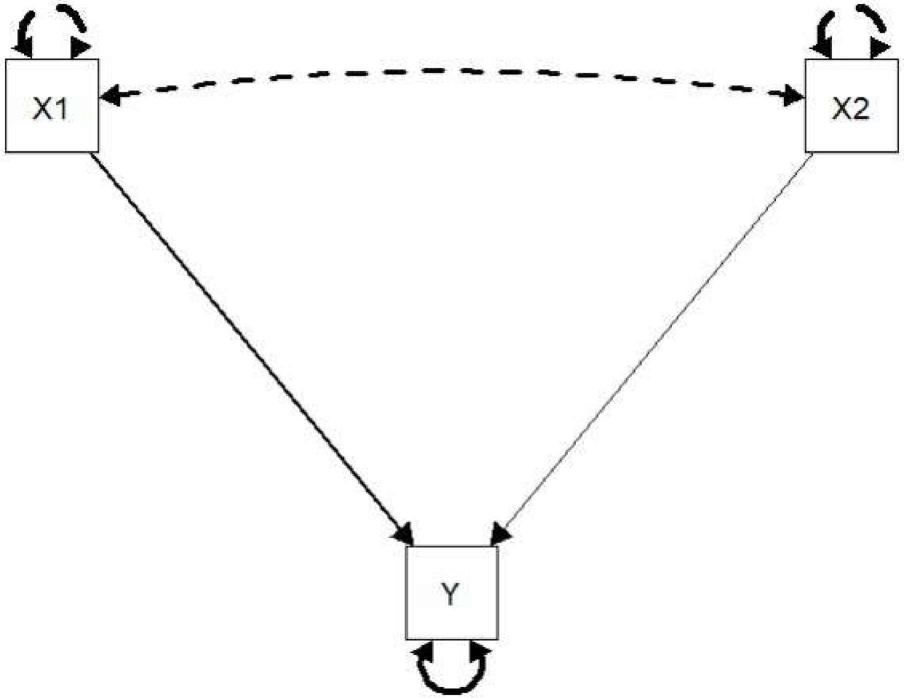
- Input:
  - Assumed model



Remark: in a SEM analysis, we estimate the coefficients for the assumed model, we do not prove that the assumed direction of the effects is correct!

SEM: An example

- Input:
  - Assumed model
  - Raw data or covariance/correlation matrix & n



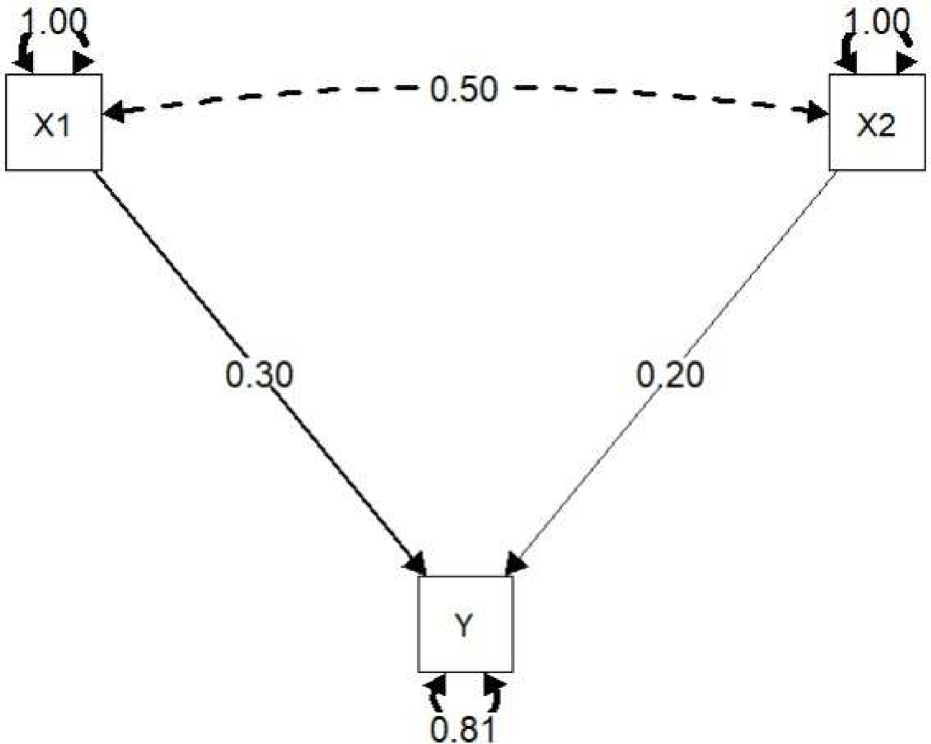
	Y	X1	X2
Y			
X1	.40		
X2	.35	.50	

n = 120

SEM: An example

- Input:
  - Assumed model
  - Raw data or cov/corr matrix
- Output:

	Y	X1	X2
Y			
X1	.40		
X2	.35	.50	



SEM: An example

- Input:

	Y	X1	X2
Y			
X1	.40		
X2	.35	.50	

- Output:

```
Regressions:
                   Estimate  Std.Err  z-value  P(>|z|)
  Y ~
    X1                0.300    0.095    3.162    0.002
    X2                0.200    0.095    2.108    0.035

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Y                 0.803    0.104    7.746    0.000
```

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MASEM

- Two-stage MASEM (Viswesvaran & Ones, 1995):
  - Stage 1: Combine data from multiple studies to estimate correlation matrix **R**
  - Stage 2: Estimate your SEM-model via estimated matrix **R**.
- One-stage MASEM (Jak & Cheung, 2020)
- Why?
  - Increase power and precision
  - Studying moderator effects on SEM parameters
  - Flexibility (latent & observed variables; MASEM model can be different from models from primary studies)

Stage 1 of MASEM

- Primary studies may vary in sample size (and therefore the precision of observed correlation may vary).
- Primary studies may report different subsets of variables (and report on different models, e.g., simple regression, multiple regression, mediation model, …)
- Pooled correlations therefore have different precision.
- Use one <u>multivariate</u> MA rather than separate MAs for each correlation, in order to account for dependence between correlations from same study e.g. r_{XY} and r_{XZ}.
- Use RVE

Stage 2 of MASEM

- Cfr. traditional SEM analysis on correlation matrix
- But: accounting for (co)variance matrix of the estimated correlations (reflecting the precisions)

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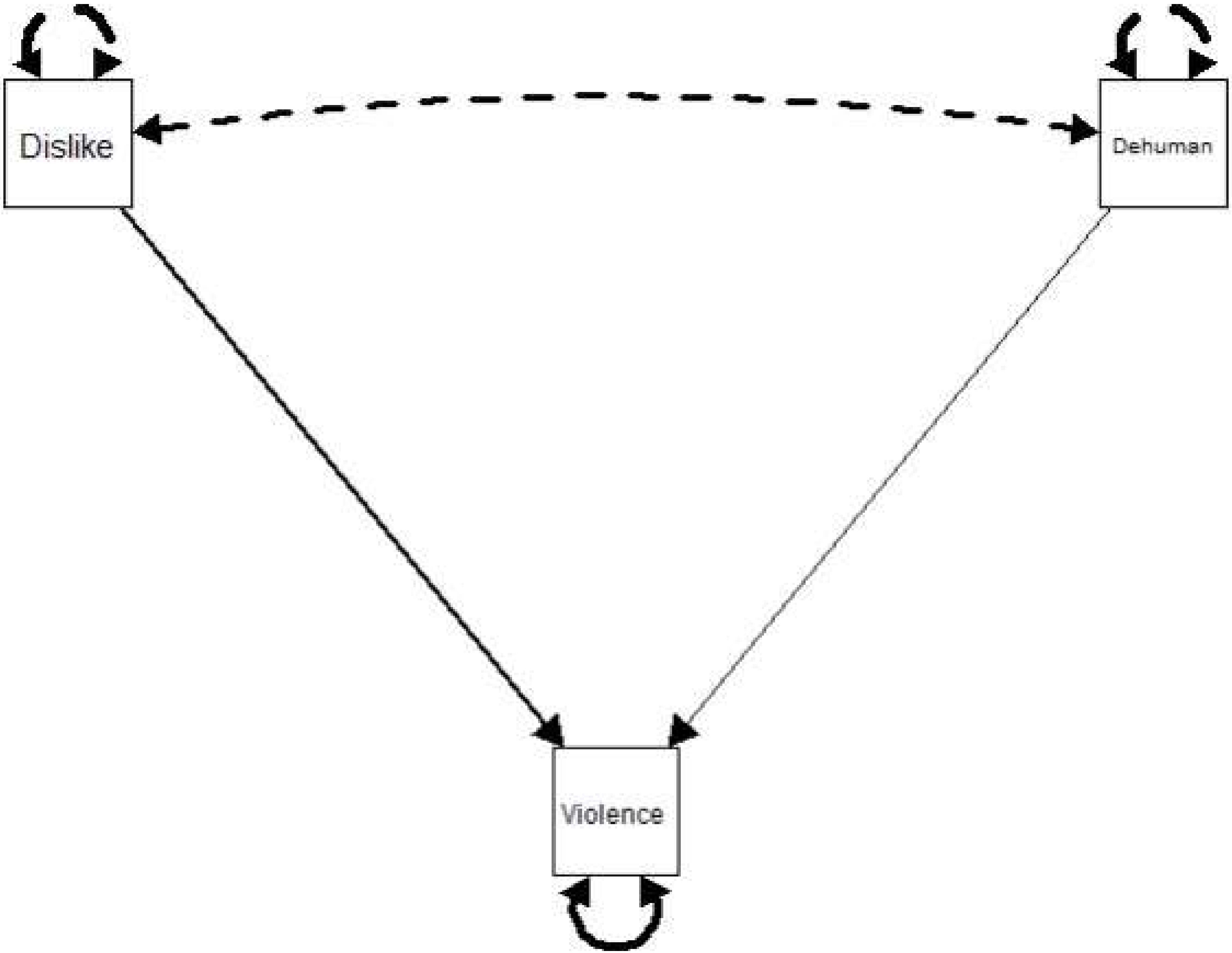
Two-stage MASEM: An example

Landry, A. P., Ghezae, I., Abou-Ismail, R., Spooner, S., August, R. J., Mair, C., Ragnhildstveit, A., Van den Noortgate, W., Gelfand, M. J., & Seli, P. (2025). The uniquely powerful impact of explicit, blatant dehumanization on support for intergroup violence. *Journal of Personality and Social Psychology.* Advance online publication. https://doi.org/10.1037/pspi0000492

Does dehumanization affect support for violence against another group, after correcting for dislike?

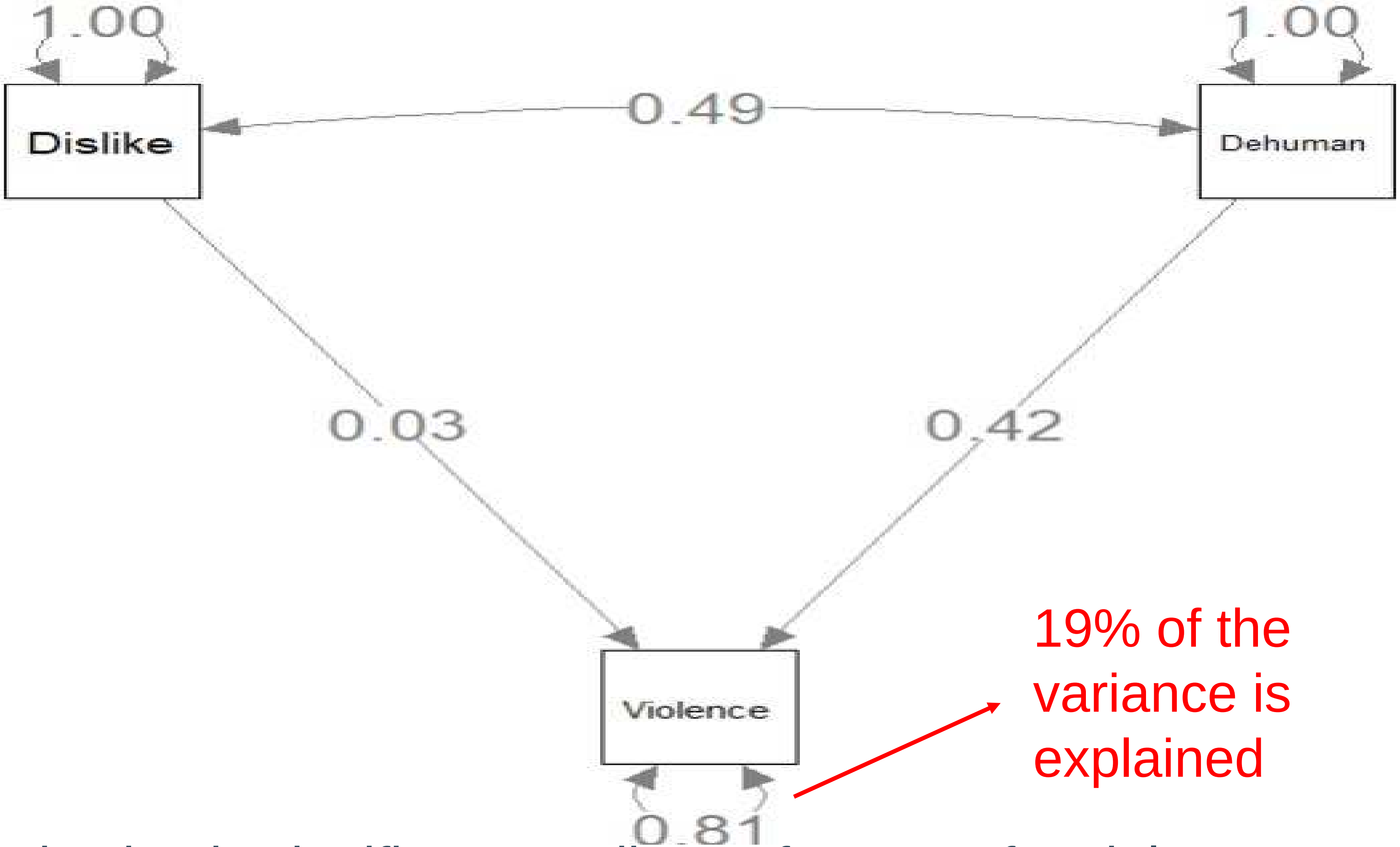

Dislike
Dehuman
Violence

Stage 1

- Multivariate MA (also applying RVE)
- Results in correlation matrix + (co)variance matrix:

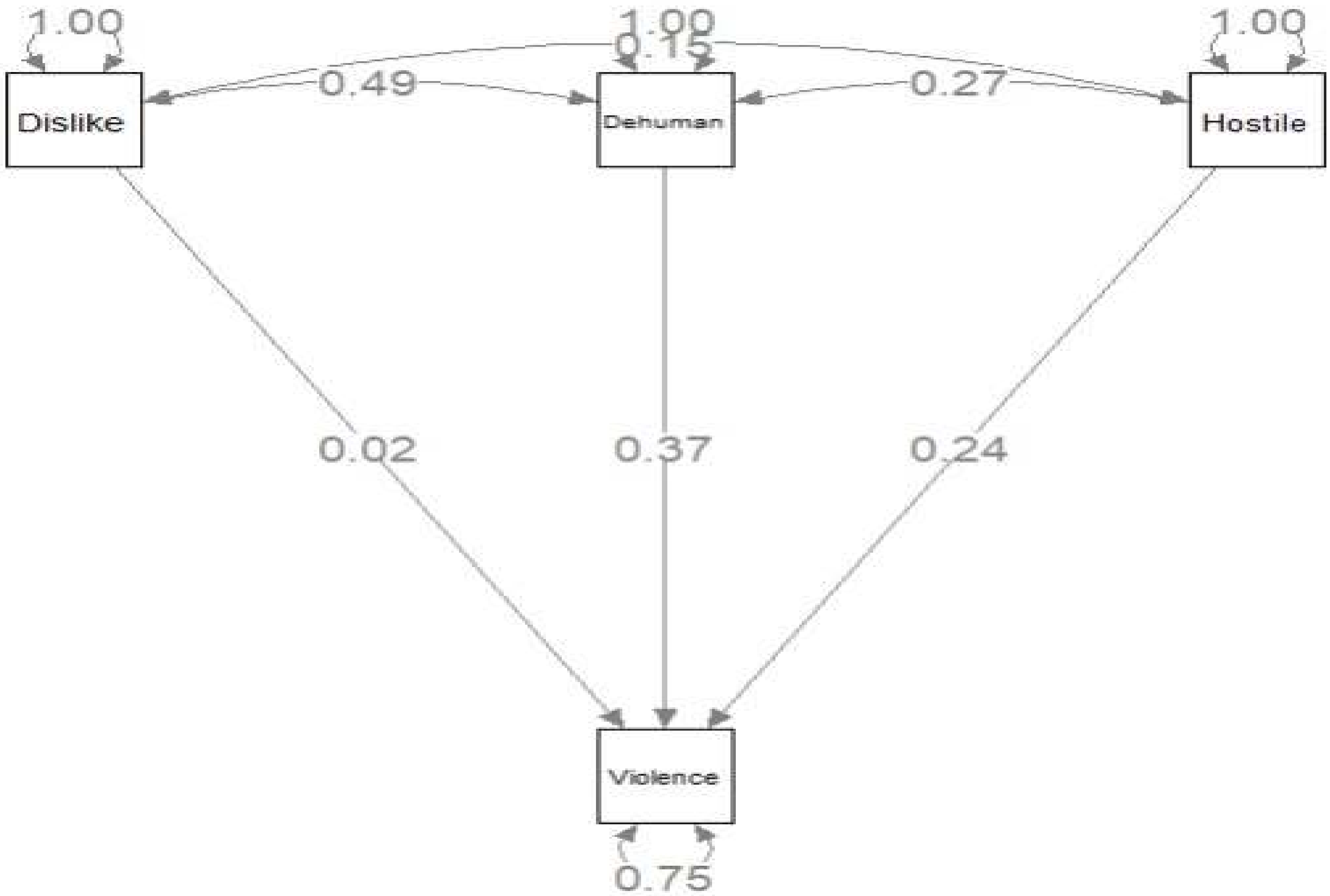
	Violence	Dislike	Dehum.
Violence			
Dislike	.24		
Dehumanization	.44	.49	

	Dislike – Violence	Dehum. - Violence	Dehum. - Dislike
Dislike - Violence	0.00176	0.00083	0.00064
Dehum. – Violence	0.00083	0.00067	0.00052
Dehum. – Dislike	0.00064	0.00052	0.00064



Dehumanization is significant predictor of support for violence over and above dislike, β = .42 [.38, .48], $\chi^2(df=1) = 336.7$, $p < .001$.
Dislike does not predict support for violence after accounting for dehumanization, β = .03 [-.05, .11], $\chi^2(df=1) = 0.53$, $p = .47$.

Sensitivity analyses showed that the association found was robust when other predictors were added



Conclusions

- MASEM is a very flexible approach: A MASEM model can contain:
  - Observed and latent variables
  - Multiple levels
  - Complex structural relations, like mediating and moderating effects
- More research is needed on possibilities and performance.

References

Cheung M.W.-L (2015). *Meta-analysis: A structural equation modeling approach*. Wiley & Sons, Inc.

Jak, S. (2015). *Meta-analytic structural equation modeling*. Springer.

Jak, S. et al. (2021). Meta-analytic structural equation modeling made easy: A tutorial and web application for one-stage MASEM. *Research Synthesis Methods, 12*, 590-606.

Jak, S. & Cheung, M. W.-L. (2020). Meta-analytic structural equation modeling with moderating effects on SEM parameters. *Psychological Methods, 25*, 430-455.